1. Introduction to SDLC

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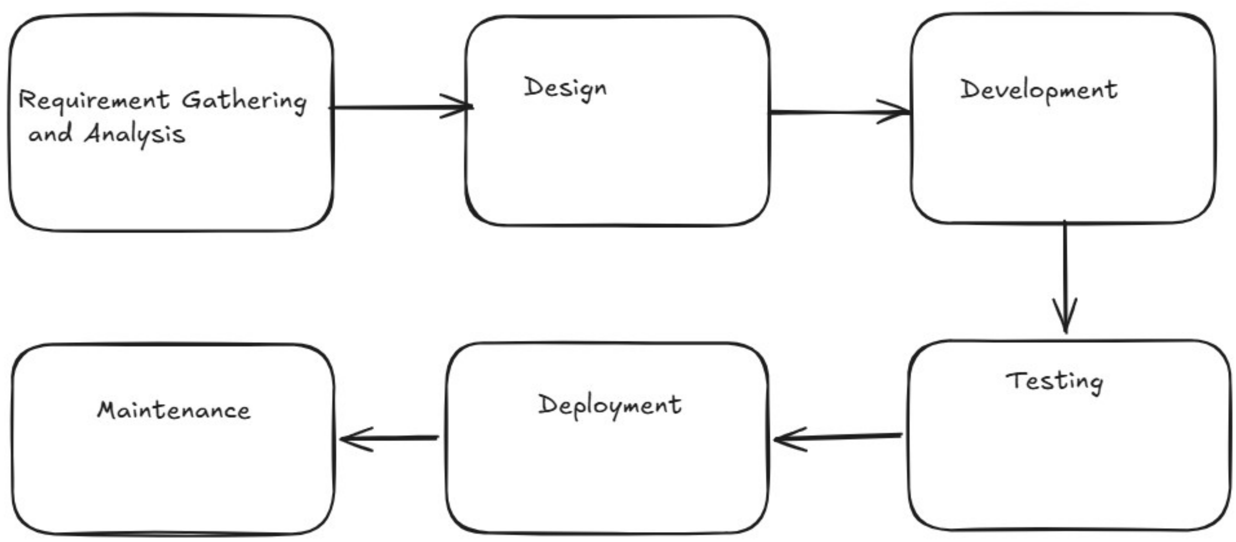
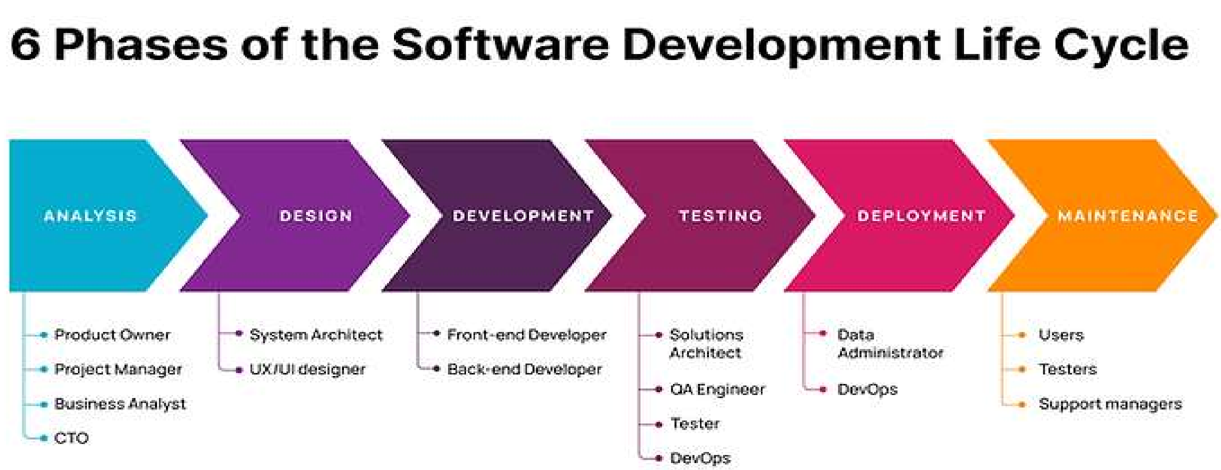
The **Software Development Life Cycle (SDLC)** is a structured framework that defines every step involved in building software—from the initial idea to the final retirement of the product. It provides a detailed, disciplined, and systematic approach to planning, creating, testing, deploying, and maintaining high-quality software.

SDLC outlines the **complete development cycle**, ensuring that all tasks across every phase are clearly defined and executed. By following SDLC, teams can deliver software that is reliable, efficient, and aligned with business requirements.

In essence, SDLC is the **end-to-end process** that guides software development through its entire lifespan: **from inception → development → testing → deployment → maintenance → retirement.**
Adhering to SDLC ensures that software is developed in a **methodical, organized, and quality-driven** manner.





SDLC Phases in Detail

1. **Requirement Gathering and Analysis** During this phase, all the relevant information is collected from the customer to develop a product as per their expectation. Any ambiguities must be resolved in this phase only

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Business analyst and Project Manager set up a meeting with the customer to gather all the information like what the customer wants to build, who will be the end-user, what is the purpose of the product. Before building a product a core understanding or knowledge of the product is very important.

For Example, A customer wants to have an application which involves money transactions. In this case, the requirement has to be clear like what kind of transactions will be done, how it will be done, in which currency it will be done, etc.
Once the requirement gathering is done, an analysis is done to check the feasibility of the development of a product. In case of any ambiguity, a call is set up for further discussion.

Once the requirement is clearly understood, the SRS (Software Requirement Specification) document is created. This document should be thoroughly understood by the developers and also should be reviewed by the customer for future reference.

2. **Design** In this phase, the requirement gathered in the SRS document is used as an input and software architecture that is used for implementing system development is derived.
   In this stage, the approved requirements are transformed into a technical blueprint for implementation.
   - **High-Level Design (HLD): Defines system architecture, technology stack, database design, and major modules.**
   - **Low-Level Design (LLD):** Specifies component logic, APIs, data structures, and workflows.
   - **Output:** Design Document Specification (DDS)

3. **Implementation or Coding** Implementation/Coding starts once the developer gets the Design document. The Software design is translated into source code. All the components of the software are implemented in this phase
   Developers build the software based on the approved design.
   - **Activities:** Coding, code reviews, unit testing, version control management
   - **Tools:** IDEs, version control systems, debuggers
   - **Output:** Source code, executable application
   - **Key Roles:** Frontend, Backend, Full Stack Developers

4. **Testing** Testing starts once the coding is complete and the modules are released for testing. In this phase, the developed software is tested thoroughly and any defects found are assigned to developers to get them fixed.

   Retesting, regression testing is done until the point at which the software is as per the customer's expectation. Testers refer SRS document to make sure that the software is as per the customer's standards.

   Testing ensures the software meets requirements and is free from defects before release.
   Types of Testing include:
   - **Unit Testing**: Verifies individual components
   - **Integration Testing**: Ensures modules work together
   - **System Testing:** Validates the complete system
   - **User Acceptance Testing (UAT)**: Confirms business requirements are met

   Output: Test cases, defect reports, and quality metrics.

5. Deployment Once the product is tested, it is deployed in the production environment or first UAT (User Acceptance testing) is done depending on the customer expectation. In the case of UAT, a replica of the production environment is created and the customer along with the developers does the testing. If the customer finds the application as expected, then sign off is provided by the customer to go live.

   The tested software is released to users.
   - **Activities:** Production setup, deployment, smoke testing
   - **Modern Approach:** Continuous Integration and Continuous Deployment (CI/CD) pipelines for faster and reliable releases
   - **Output:** Live application
   - **Key Roles:** DevOps Engineers, Release Managers

6. Maintenance After the deployment of a product on the production environment, maintenance of the product i.e. if any issue comes up and needs to be fixed or any enhancement is to be done is taken care by the developers.

   Post-deployment support ensures long-term usability.
   - **Activities:** Bug fixes, performance tuning, updates, feature enhancements
   - **Output:** Patches, updates, new versions
   - **Key Roles:** Support Engineers, Developers